

KARTIKEY AGARWAL

+91-8392822445 | Kartikeyeagarwal234@gmail.com | [LinkedIn](#) | [GitHub](#)

Education

Vellore Institute of Technology

B.Tech in Computer Science and Engineering

Sep 2022 – May 2026

CGPA: 7.91/10

Meerut Public School

Class XII (CBSE)

Apr 2021 – Mar 2022

95%

Meerut Public School

Class X (CBSE)

Apr 2019 – Mar 2020

90%

Technical Skills

Languages: Swift

iOS Frameworks: UIKit, SwiftUI, Foundation

Backend & Storage: Firebase Authentication, Cloud Firestore, Core Data, SwiftData

Architecture: MVC, MVVM, Delegate Pattern

Networking: URLSession, REST APIs

Projects

TripMind | *Swift, SwiftUI, Core Data, Gemini AI* | [GitHub](#)

- Developed a full-featured **iOS travel planning app** with AI-generated packing lists, trip management, and real-time packing progress tracking
- Integrated **Google Gemini AI** via **URLSession** and **async/await** to deliver instant, context-aware packing suggestions from plain-language trip descriptions
- Implemented **Core Data** persistence with full **CRUD** operations, ensuring all trips and packing items survive app restarts and background states
- Adopted **MVVM** architecture with a clean **TripViewModel** layer, decoupling business logic from the UI for maintainability and testability
- Enforced **secure API key management** by keeping secrets out of version control via a gitignored **Secrets.swift** file

MailDigest | *Swift, SwiftUI, SwiftData, Groq API (Llama 3.1)* | [GitHub](#)

- Built an **AI-powered email digest app** that fetches Gmail messages and generates one-line LLM summaries using **Groq's Llama 3.1**, reducing email-reading time significantly
- Implemented **OAuth 2.0** authentication via **Google Sign-In SDK** and consumed the **Gmail REST API** for secure, read-only inbox access
- Engineered **automatic email categorization** (Important / Promotions) and full-text search across senders, subjects, and AI summaries for faster triage
- Leveraged **SwiftData** (iOS 17+) for offline-capable local persistence and built swipe-to-delete and read/unread toggle interactions
- Designed **AI-driven push notifications** surfacing LLM-generated insights instead of raw subject lines, delivering smarter and more actionable alerts

Experience

Research Intern

Dec 2024 – Jan 2025

Indian Space Lab

- Conducted research on UAV technologies for astronomy applications, focusing on performance metrics and sensor systems
- Evaluated multiple UAV configurations for payload capacity, stability, and data collection accuracy
- Structured and analyzed telemetry data from test flights to support research recommendations

Leadership & Achievements

Core Member, AI Club — Organized technical workshops on ML fundamentals for 200+ students

Amazon HackOn Participant — Participated in national-level hackathon

National-Level Taekwondo Athlete — Demonstrated discipline, perseverance, and competitive excellence